

Beyond Maximal Contrast: A Literature Review on The Effect of Text-Background Colour, Display Polarity, and Visual Fatigue on Digital Displays

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Abstract—Digital displays are commonly designed under the assumption that maximum luminance contrast, such as black text on a white background, provides the best conditions for reading, and the widespread adoption of dark mode has made that assumption more visible rather than less. This paper reviews literature on the relationship between contrast and readability, and asks whether contrast alone is sufficient to predicting reading performance and visual comfort on digital displays, and examines other key factors that may be overlooked for the obvious conclusion. Thirteen empirical studies and the Web Content Accessibility Guidelines provided by the World Wide Web Consortium (W3C) are examined across five themes: luminance contrast, display polarity, text and background hue, ambient illumination, and user and task characteristics. The review finds that text must meet a minimum level of contrast to be legible, and that higher contrast generally improves character recognition and short-term legibility; however, the highest available contrast is not always the most effective choice. Positive polarity is advantageous for proofreading, visual acuity, and glance-like reading, whereas negative polarity can reduce objective measures of visual fatigue during prolonged reading in very dark surroundings. The influence of hue is typically smaller than that of luminance, yet particular combinations measurably improve or degrade recognition, attention, preference, and fatigue: warm backgrounds increase reading speed for readers with and without dyslexia, and yellow text on a black background reduces eye strain under negative polarity. Finally, performance, comprehension, fatigue, and preference frequently diverge, so no single colour combination is superior on every measure. Contrast, therefore, is best treated as a necessary baseline rather than as a design goal in itself; designers should select colours according to reading duration, ambient lighting, and device, and should allow users to adjust these settings. The review closes by proposing a sensor-instrumented smartphone experiment designed to resolve the sampling, calibration, and single-outcome limitations that recur across the reviewed work.

I. INTRODUCTION

Reading on a digital display has become ubiquitous. Educational materials, workplace documents, news, transport systems, wearable devices, and commercial signage all use self-illuminating screens to deliver information. Design decisions about how text and background colours are differentiated therefore affect the readability and accessibility of digital content for a wide range of users. The most common design rule is to maximize contrast. This convention is reinforced by the Web Content Accessibility Guidelines (WCAG) 2.1, which specify a contrast ratio of at least 4.5:1 for normal-sized text and 3:1 for large text at Level AA [13]. Such requirements

are essential for preventing information from becoming inaccessible to people with reduced vision. A minimum standard is not an optimum, however, and contrast ratios well above the threshold do not necessarily indicate better readability or a more appropriate design choice.

The question motivating this survey is as follows: *is contrast the single most important factor for text readability, particularly on a digital reading device, or can other colour combinations that do not maximize contrast—that is, combinations other than pure black-on-white or white-on-black—produce equal or better reading performance and lower visual fatigue, depending on the reading task, ambient lighting, and environmental conditions?*

The intended audience includes accessibility specialists, user-interface and web designers, e-reader and mobile-device manufacturers, standards bodies whose guidance is contrast-centred, and researchers in human-computer interaction and visual ergonomics. The question matters most for people who read for prolonged periods, use screens at night, experience age-related visual change, or have dyslexia, since these groups are the most sensitive to display design decisions.

Existing work indicates that warm background colours improve reading performance for readers both with and without dyslexia [11], and that display polarity affects legibility for younger and older adults alike [10]. Small, low-contrast text has also been shown to be disproportionately harmful to readers over 40, producing eyestrain at significantly higher rates [14]. A clearer answer would therefore improve the reading experience of many users, particularly older readers of digital interfaces and anyone who reads on a screen for long stretches at night or in dim lighting.

II. BACKGROUND

Research on text and background colour combinations dates back to studies of cathode ray tube (CRT) displays. In an early study, Pastoor [9] evaluated hundreds of foreground-background colour combinations and found that, once contrast was adequate, chromaticity—the particular hues chosen—had little effect on text legibility, although it did affect subjective preference. Results of this kind established the convention that luminance contrast governs legibility while hue governs taste, and later work on web pages [5], LCD displays [6], and negative-polarity interfaces [8] largely reinforced it. The same

logic underpins contrast-based standards such as WCAG [13], in which a minimum recommended contrast ratio serves as the operational definition of readable text.

Three developments complicate that convention. First, display technology has changed: legibility rankings derived from paper and CRT screens do not transfer reliably to LCD panels [6], and OLED devices with true black pixels and per-pixel emission differ again. Second, black-on-white and white-on-black have identical WCAG contrast ratios of 21:1 and yet are demonstrably not equivalent for readers, which means a directionless ratio cannot express something the eye clearly registers. Third, the outcomes researchers care about have multiplied beyond character recognition to include comprehension, physiological fatigue, attention, and preference, and these outcomes do not move together. The remainder of this paper examines how contrast, polarity, and hue behave once those three complications are taken seriously.

III. LITERATURE SURVEY

A. Survey Scope and Analytical Approach

This survey examines thirteen empirical studies together with the Web Content Accessibility Guidelines, supplemented by earlier CRT-era work [9] used for historical context. The studies investigate text and background colour across several classes of digital display, including web pages, desktop monitors, LCD screens, smartwatches, and experimentally controlled reading displays. The tasks used include character identification, proofreading, visual search, connected-text reading, lexical decision, and comprehension or retention testing. Reported outcomes are similarly varied, spanning recognition accuracy, reading speed, visual fatigue, physiological response, aesthetic judgment, and user preference. The survey is narrative rather than systematic: studies were selected for their relevance to contrast, polarity, and hue on self-illuminating displays rather than through an exhaustive protocol-driven search.

The evidence cannot be compared through a simple count of which studies favour which colour combination. Some experiments isolate a single factor, such as brightness or polarity, while holding other variables constant; others deliberately compare complete colour schemes in which hue, saturation, and luminance vary together. Exposure duration also differs considerably, from recognition during presentations lasting milliseconds to fatigue accumulated over half an hour. Ambient lighting, display technology, font size, participant age, and task type vary as well.

For these reasons, the survey adopts a context-based approach to interpretation. Each result is considered in relation to the outcome measured, the task performed, the lighting and display conditions, and the population studied. This approach helps explain why apparently conflicting findings may both be valid within their own conditions. Rather than attempting to identify one universally superior colour combination, the survey seeks to establish the conditions under which contrast, polarity, and hue become more or less important.

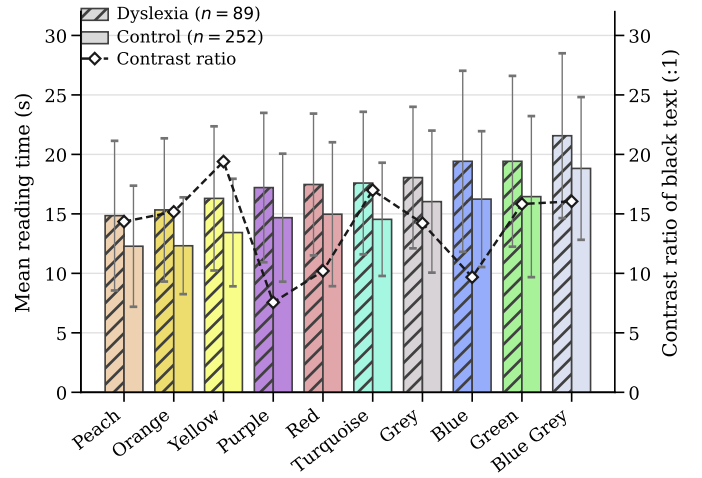


Figure 1. Mean reading time for black text on ten background colours, for readers with and without dyslexia, from Rello and Bigham [11]. Bars are filled with the background colour tested; error bars show one standard deviation. The dashed series gives the WCAG luminance contrast ratio of black text against each background. Every background exceeds the 7:1 AAA threshold, yet reading times span 45% for readers with dyslexia and 53% for controls. The ordering of contrast ratios bears no relation to the ordering of reading times: the highest-contrast background (yellow, 19.4:1) was not the fastest, and the third-highest (blue grey, 16.1:1) was the slowest. The two groups' rankings are almost perfectly correlated ($\rho = 0.964$).

B. Core Concepts: Contrast, Polarity, Hue, and Saturation

Luminance contrast represents the difference in brightness between text and its background. WCAG defines it as a ratio based on the relative luminance of two colours, whereas laboratory studies more often use Michelson contrast, the normalized difference between maximum and minimum luminance. Under either definition, a value close to zero means that the foreground is nearly indistinguishable from the background. Accordingly, the literature agrees that very low contrast should be avoided [13, 8, 14, 15].

Polarity describes the direction of the luminance relationship between text and background. Positive polarity refers to dark text on a light background, such as black on white; negative polarity refers to light text on a dark background, such as white on black. The two arrangements may have identical contrast ratios and yet produce different outcomes. A bright background constricts the pupil, which can reduce optical aberration and increase depth of focus, whereas a dark background reduces the total light reaching the retina and is generally more comfortable in dim surroundings.

Hue and saturation describe the chromatic properties of a colour independently of how light or dark it is. Studies that manipulate hue must control luminance, because an apparent effect of hue may otherwise be an effect of brightness. Li *et al.* [8], for example, matched their background colours on brightness and saturation for precisely this reason.

C. Contrast Alone is Not Sufficient

The reviewed literature agrees that insufficient contrast harms legibility. The WCAG minimum contrast ratio establishes a floor that prevents text from becoming unreadable [13].

Empirical work supports this floor. Wang *et al.* found that low contrast reduced the legibility of text on smartwatches and that increasing contrast partly compensated for small text, particularly for users over 40 [14]. Li *et al.* likewise documented that increasing brightness contrast improved recognition under negative polarity [8]. Xie *et al.* showed that higher Michelson contrast reduced visual fatigue during night-time reading, with the lowest contrast level producing the greatest eye strain [15]. Adequate contrast is thus a precondition for both readability and visual comfort.

The relationship between contrast and readability is not, however, a matter of simply increasing contrast without limit. Ko compared four levels of luminance contrast in a brand-icon recognition task and found that the highest level, 18:1, outperformed the others, while the three lower levels produced broadly similar results [7]. Because those three levels did not differ appreciably, the relationship between contrast and recognition appears non-linear rather than uniformly increasing. Hall and Hanna reported a related pattern: although participants rated black text on a white background as the easiest to read, retention of the page content did not differ significantly across colour combinations [5]. Once an adequate contrast level has been reached, further increases may improve the reading experience without improving retention.

The research also shows that chromatic contrast cannot substitute for luminance contrast. Buchner and Baumgartner created strong colour differences using red and green while holding luminance contrast very low; despite the pronounced chromatic difference, proofreading performance remained poor [1]. Colours that appear clearly distinct may therefore still be difficult to read when their brightness values are too similar.

Luminance alone does not determine which high-contrast combinations perform best. Ko found that several combinations with similar contrast ratios produced noticeably different levels of recognition: yellow-on-black, yellow-on-blue, and white-on-blue performed best, while other combinations at comparable ratios performed considerably worse [7]. Humar *et al.*, after evaluating 56 colour combinations on LCD displays, similarly found that legibility rankings developed for paper or older CRT monitors did not transfer reliably to modern LCD screens [6]. The clearest demonstration comes from Rello and Bigham [11], whose ten background colours all exceeded the strictest WCAG threshold and yet produced reading times differing by 45–53%, with no correspondence between a background’s contrast ratio and how quickly it was read (Fig. 1).

Once the minimum contrast requirement has been met, therefore, choosing an effective design becomes an optimization problem that balances contrast against colour, polarity, device type, user comfort, and the intended reading task.

D. Display Polarity: Equal Contrast Does Not Mean Equal Performance

Most research indicates that positive polarity is strongest when readers are proofreading, reading small text, or recognizing words quickly. Buchner and Baumgartner conducted four

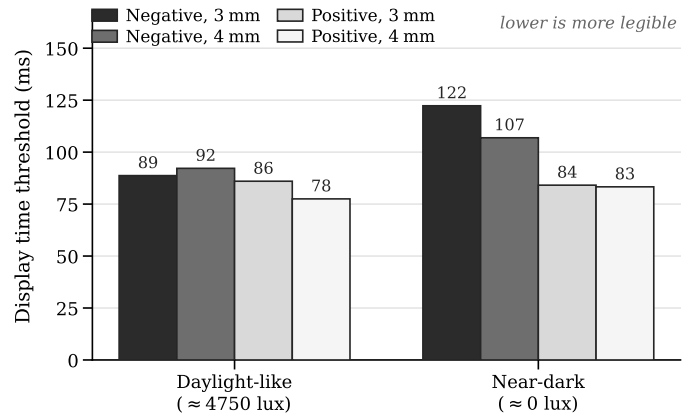


Figure 2. Glance legibility thresholds reported by Dobres, Chahine and Reimer [2]: the mean on-screen display time required for approximately 80% accuracy in a lexical decision task, for two contrast polarities and two letter sizes under two ambient lighting conditions. Lower thresholds indicate greater legibility. Under daylight-like illumination the two polarities are indistinguishable, whereas in near-darkness negative polarity requires roughly 45% longer exposure at the smaller letter size—a significant polarity by illumination interaction ($F(1,33) = 14.18, p < 0.001$). Error bars are omitted because the source reports between-subject standard deviations while the analysis was within-subject.

experiments using achromatic and chromatic displays under both low and office-like illumination, and proofreading was consistently better with dark text on a light background [1]. The advantage did not depend on ambient illumination or on whether the colours were black and white or blue and yellow. Physiological and self-report measures gave no indication that participants exerted greater effort in the more difficult conditions, so the result reflects a genuine performance difference rather than a performance-effort trade-off.

Piepenbrock *et al.* examined whether the positive-polarity advantage extends across age, using a large between-subjects experiment with younger and older adults. Positive polarity improved both visual acuity and proofreading for both groups; age reduced performance overall and moderated the size of the polarity effect, but did not reverse its direction [10]. This suggests that the greater retinal illuminance of positive-polarity displays offsets some age-related losses in readers with normal or corrected-to-normal vision.

Although black-on-white and white-on-black have equal contrast ratios, then, positive polarity generally outperforms negative polarity across ages and lighting conditions when contrast is held constant.

E. Ambient Illumination, Visual Fatigue, and Task Context

Although positive polarity generally produces better reading performance, the advantage is not constant across situations. Ambient lighting, task type, and reading duration all influence the size of the polarity effect and, in some cases, whether dark mode offers a practical benefit.

Dobres *et al.* showed that ambient illumination modulates the polarity effect in glance-like reading. Participants made lexical decisions about briefly presented words at two font sizes under dark and daylight-like illumination. As Fig. 2

shows, negative-polarity thresholds were substantially worse in the dark condition, whereas bright ambient illumination improved all conditions and effectively eliminated the polarity gap [2]. The authors therefore suggest that the effect is better described as a negative-polarity disadvantage than as a positive-polarity advantage: dark mode is not simply less good in the dark, it is specifically penalized there. Using dark mode because the surroundings are dark is consequently a poor choice when the task is rapid recognition rather than sustained reading.

Task context also shapes outcomes at very short exposures. Teixeira *et al.* found that during fifteen-second reading exposures, yellow backgrounds attracted initial attention fastest but supported the slowest sustained processing, whereas black backgrounds captured attention more slowly yet required the shortest total viewing time [12].

To approximate device use after a room's lights have been switched off, Xie *et al.* studied reading at less than 3 lux with low screen luminance. In this context, negative polarity produced lower objectively measured visual fatigue, particularly at high contrast, even though participants preferred positive polarity [15]. Positive polarity is thus better suited to improving acuity and proofreading accuracy, while negative polarity can reduce emitted light and physiological strain during sustained use in a dark room.

Positive polarity is consequently the safer default for general-purpose text, proofreading, small text, older readers, and glance-like tasks. Negative polarity becomes preferable when ambient light and screen luminance are very low, when the task involves sustained rather than glance-based reading, and when the design avoids text colours or contrast combinations that impede reading or provoke eye strain.

F. Hue and Specific Colour Combinations

Hue usually exerts a smaller effect than luminance contrast, and its apparent influence depends heavily on how luminance, saturation, and polarity are controlled. It is not, however, irrelevant.

Under negative polarity, Li *et al.* compared six background colours matched on brightness and saturation while varying text and background brightness. Higher brightness improved legibility, and brighter text produced higher ratings of legibility, appearance, and comfort, but background hue produced no significant effect [8]. When brightness is carefully controlled, hue alone therefore appears to have only limited influence on readability.

Not all studies examined hue in isolation. Guo *et al.* investigated complete colour schemes, varying hue, saturation, brightness, text colour, and background texture across their tests. They found that green-on-black and green-on-white produced the strongest recognition performance, whereas peach and light grey performed worst, and background texture had little effect on recognition efficiency [4]. Because several visual properties changed at once, the observed improvements cannot be attributed to hue alone. The findings, nonetheless,

suggest that some complete palettes outperform others when multiple design elements interact.

Within negative polarity, Fan *et al.* found that red text caused the greatest visual fatigue, followed by green, blue, white, and yellow [3]. Yellow and white, however, supported the strongest cognitive performance. Even among high-contrast colours, light text colours are not interchangeable, and designers should not assume that any sufficiently bright colour will perform equally well in a dark-mode interface.

To examine the effect of warm backgrounds, Rello and Bigham studied 341 readers, 89 of whom had dyslexia. As Fig. 1 shows, black text on warm peach, orange, and yellow backgrounds produced faster reading and less mouse movement than cool blue-grey, blue, and green backgrounds [11]. The improvement appeared for readers both with and without dyslexia, indicating that the benefit of warm backgrounds is not restricted to readers with reading disabilities. The figure also isolates the survey's central claim in a single dataset: because every background comfortably exceeded the strictest WCAG threshold, contrast ratio has no remaining explanatory power over the resulting spread in reading times.

Similarly, Ko found that yellow-on-black and yellow-on-blue were among the highest-ranked combinations for legibility in brief exposures, suggesting that certain pairings are especially effective for short labels and icons [7]. Teixeira *et al.*, by contrast, found that yellow backgrounds attracted attention quickly but slowed sustained processing during longer reading [12]. Colours optimized for attracting attention are thus not necessarily suited to prolonged reading.

Overall, hue influences readability less than luminance contrast or display polarity when brightness is carefully controlled. Once an appropriate level of contrast has been reached, however, colour combination can still affect recognition, visual fatigue, aesthetic judgment, and reading speed. Designers should therefore select colours according to the task being performed and the experience they intend to create, rather than applying a single scheme to every context.

G. Divergent Outcomes: Hall and Hanna as a Case Study

A conclusion about colour depends heavily on which outcome is measured, and Hall and Hanna [5] is the clearest illustration because it is the only reviewed study to measure perceived readability, retention, aesthetic response, and behavioural intention within a single design. In a between-subjects experiment with 136 participants, four text-background combinations varying in both contrast and chromaticity were applied to an educational page and a commercial page, with layout and content standardized across conditions.

The four outcomes dissociated. Greater contrast improved perceived readability; colour combination had no significant effect on retention; and chromatic schemes, particularly blues, raised both aesthetic ratings and purchase intention. Black-on-white therefore won on exactly one measure, was indistinguishable on a second, and lost on the remaining two. A designer following a contrast-maximizing rule would optimize the out-

come on which the schemes differed least while sacrificing those most relevant to a commercial interface.

Several constraints qualify this. Only four colour conditions were tested, so the shape of the contrast-outcome relationship cannot be characterized—a limitation shared with Ko [7]. Monitor luminance was not controlled, a compromise the authors defend on the grounds that real displays also vary. The session was short and single-sitting, so nothing follows about fatigue during prolonged reading, which is where the polarity and hue literature locates the strongest effects [15, 3]. Finally, the retention result is a null finding reported without an equivalence test or power analysis, and so establishes only that no difference was detected. Notwithstanding these constraints, measuring four outcome classes concurrently is what made the dissociation visible at all, and it is the approach this survey recommends for future work.

IV. LIMITATIONS

Although several consistent trends emerged across the reviewed studies, the findings should be interpreted within the constraints of the available literature. The experiments used different display technologies, reading tasks, typography, ambient lighting conditions, and outcome measures, which makes direct comparison difficult. These methodological differences broaden the range of situations investigated, but they also limit the strength of any universal recommendation drawn from a single study.

A further limitation concerns the composition of the samples. Many studies recruited university students or younger adults with normal or corrected-to-normal vision. Although research involving older adults and readers with dyslexia generally supported the same overall trends, comparatively little evidence exists for children, individuals with low vision, or people with colour-vision deficiencies. The conclusions presented here may therefore not generalize equally well to every user population.

Finally, this survey is narrative rather than systematic. Studies were selected for topical relevance rather than through a pre-registered search protocol with explicit inclusion and exclusion criteria, so the possibility of selection bias cannot be excluded, and no formal quality appraisal or effect-size synthesis was performed.

V. FUTURE WORK

A. Recurring Methodological Gaps

Five weaknesses recur across the reviewed studies and jointly explain why the literature has not converged.

Uncontrolled or inconsistently reported display conditions. Some studies calibrate screen luminance with a photometer [15, 3]; others acknowledge that monitor luminance was not controlled [5], or were conducted remotely on participants' own hardware, where display characteristics were neither set nor recorded [11]. Because the positive-polarity advantage is understood to follow from display luminance rather than from polarity as such, an uncalibrated screen leaves the principal

causal variable unmeasured, and a result obtained at one screen luminance cannot be assumed to hold at another.

Narrow and clustered ambient lighting. Studies concentrate below 10 lux [15, 3] or contrast near-darkness with typical office lighting [1], and Dobres, Chahine and Reimer [2] is the only reviewed work reaching daylight intensity, at approximately 4750 lux. The colour temperature of the ambient light is almost never manipulated at all, despite chromatic adaptation altering how screen hues are perceived.

Unrepresentative and under-characterized samples. University students and young adults predominate [4, 12, 3]. Even the largest sample was 83% degree-holding [11]. Education, technology familiarity, and habitual screen use are essentially never recorded, so they can be neither controlled nor tested as moderators; Xie et al. [15] attribute their preference result to participants' usage habits without having measured them.

Single-outcome designs. Most studies measure performance or fatigue or preference, but rarely all three, which forces divergences between them to be inferred across studies with incompatible methods rather than observed within one.

Devices that people no longer read on. The corpus is dominated by desktop monitors and LCD panels. Wang et al. [14] is the only study using a device carried on the body, and Dobres, Chahine and Reimer [2] explicitly motivate their work by smartphone glance reading while testing a desktop display.

B. A Proposed Instrumented Smartphone Experiment

We therefore propose an experiment that treats the smartphone not merely as the display under test—justified because it is now the device on which most reading occurs—but as the measuring instrument. Modern handsets already contain an ambient light sensor, a front-facing camera, and APIs exposing screen dimensions, pixel density, rendered text size, brightness, and night-mode state. Together these permit continuous logging of precisely the variables the reviewed studies left uncontrolled, at a sample size no laboratory could reach.

The design has two tiers. A *calibrated laboratory tier* fixes the device model, verifies screen luminance with a spot photometer at every condition, and crosses ambient illuminance (0, 10, 300, and approximately 5000 lux) with correlated colour temperature (2700 K, 4000 K, 6500 K) using tunable-white luminaires. Lighting conditions are administered on separate days in a Latin square, correcting the counterbalancing failure that Dobres, Chahine and Reimer [2] acknowledge in their own design. A *field tier* deploys the same application on participants' own devices, using the sensor stream to reconstruct viewing conditions in situ. The laboratory tier establishes internal validity; the field tier supplies the ecological validity and scale of Rello and Bigham [11] without inheriting that study's total absence of display control.

Three design choices distinguish the proposal from the reviewed work. First, colour conditions are generated in a perceptually uniform space such as CIELAB rather than in HSB, so that hue, chroma, and lightness can be varied orthogonally; this separates the hue effects that Guo et al. [4] confounded with three other properties. Second, luminance contrast is sampled

Table I
PROPOSED INSTRUMENTATION: VARIABLES COLLECTED, CAPTURE METHOD, AND RATIONALE

Category	Variable	Capture Method	Purpose and Gap Addressed
Manipulated	Text–background pair	Rendered in CIELAB at fixed lightness steps	Separates hue and chroma from luminance; Guo et al. [4] varied five properties at once
	Luminance contrast	Six or more levels, photometer-verified; recorded as both WCAG and Michelson	Permits the contrast-performance curve to be modelled as a shape; Ko [7] and Hall and Hanna [5] had only four points
	Display polarity	Application theme at matched measured luminance	Tests whether the polarity effect survives on an OLED handset
	Reading duration	Glance, 2-minute, and 25-minute blocks per participant	Measures the glance/sustained divergence within one design rather than across studies
Environment (controlled and logged)	Ambient illuminance	Tunable luminaires (lab); phone light sensor at 1 Hz (field); 0–5000 lux	Existing work clusters below 10 lux or stops at office lighting [1, 15, 3]
	Ambient colour temperature	Tunable-white luminaires; white-balance estimate from front camera	Almost never manipulated; Fan et al. [3] held it fixed at 5100 K
	Screen luminance	Spot photometer per condition; auto-brightness disabled	Displays were uncontrolled in web-based and remote studies [5, 11]; a result at one luminance does not transfer to another
	Night mode / blue-light filter	Operating-system API state, set per condition	Ubiquitous in practice, yet absent as a factor from every reviewed study
Device (logged)	Screen size, resolution, pixel density	Device API	Converts nominal point size into physical size
	Rendered font size and viewing distance	Typographic API plus front-camera face-distance estimate	Yields visual angle as the analysis unit, making sizes comparable across devices
	Panel technology	Device model lookup	CRT-era rankings did not transfer to LCD [6]; OLED may differ again
	Thermal and battery state	System API	OLED brightness throttling silently alters luminance mid-session
Participant	Age, stratified into bands	Screening questionnaire	Polarity effects vary with age [10]; samples are dominated by students
	Visual acuity and colour vision	logMAR chart and Ishihara plates	Screened inconsistently across the reviewed studies
	Education and technology familiarity	Questionnaire plus logged weekly screen time	Recorded by no reviewed study; the sample of Rello and Bigham [11] was 83% degree-holding
	Dyslexia and visual-stress screening	Validated screener	Allows the warm-background effect [11] to be tested alongside all other factors
Outcome	Sleep and caffeine in prior 24 h; time of day	Self-report and timestamp	Established fatigue confounds, unrecorded in the fatigue studies
	Reading speed, lexical decision threshold, proofreading errors	In-application timing	Task performance
	Comprehension and retention	Post-block test with equivalence bounds	Makes a null result interpretable, unlike Hall and Hanna [5]
	Blink rate and pupil diameter	Front-camera eye tracking (field); dedicated eye tracker (lab)	Objective fatigue, following Xie et al. [15] and Fan et al. [3]
	Comfort, aesthetics, and preference	Likert scales after each block	Preference and performance diverge [15, 5]

at six or more levels and modelled as a continuous predictor, allowing the shape of the contrast-performance function to be estimated rather than inferred from the four points available to Ko [7] and Hall and Hanna [5]. Third, every participant completes glance, short, and sustained reading blocks on the same stimuli, so that the divergence between glance-optimal and fatigue-optimal designs is measured within one experiment instead of across three.

All four outcome families are collected concurrently, following the multi-outcome logic of Hall and Hanna [5]: task performance, comprehension, objective and subjective fatigue, and preference. Analysis uses linear mixed-effects models with participant-level random effects, with the protocol, sample size, and equivalence bounds pre-registered so that null results—such as the retention finding of Hall and Hanna [5]—can be interpreted rather than merely reported. Table I sets out the full instrumentation.

Because rendered text size, pixel density, and viewing distance are all logged, stimulus size can be expressed as visual

angle rather than as points or millimetres. This alone would make results directly comparable across devices and across studies, which the present literature is not.

C. Further Directions

Beyond this experiment, three questions remain open. A direct test of the moderate-contrast hypothesis is still absent: no reviewed study compared a moderate-contrast, warm, positive-polarity scheme—near-black text on a warm off-white background—against maximum-contrast black-on-white across a sustained session while recording speed, comprehension, and fatigue together. Additional work is also needed on display technologies whose optical properties differ from the LCD panels used in most existing research, including mini-LED, high-refresh-rate, and electronic paper devices. Finally, broader participant populations should be recruited, particularly children, older adults, individuals with low vision, and readers with colour-vision deficiencies, since a better understanding of how these groups respond to colour and

contrast is what would allow accessibility guidance to extend beyond a minimum contrast ratio.

VI. CONCLUSION

The objective of this survey was to determine whether maximizing text-background contrast is by itself sufficient to produce the best reading performance and visual comfort on digital displays. Across the thirteen empirical studies reviewed, the evidence consistently shows that adequate luminance contrast is essential for readability, supporting the minimum recommendations established by the Web Content Accessibility Guidelines. Very low contrast reduces recognition accuracy, increases visual fatigue, and disproportionately affects users reading small text as well as older adults.

Once an appropriate level of contrast has been achieved, however, readability depends on considerably more than contrast. Display polarity, ambient illumination, reading duration, device characteristics, and the chosen colour combination all influence reading performance. Positive polarity generally provides superior proofreading performance, visual acuity, and recognition accuracy, whereas negative polarity can reduce physiological measures of visual fatigue during prolonged reading in very dark environments. Hue typically has a smaller effect than luminance contrast, yet particular combinations improve recognition speed, reduce fatigue, or increase user preference depending on the task. Readability also cannot be described by any single outcome measure: performance, physiological response, comfort, and preference diverge often enough that the “best” colour combination depends on the goals of the interface.

The reviewed evidence provides only partial support for the expectation that moderate-contrast, warm, positive-polarity schemes can match maximum-contrast black-on-white. Warm backgrounds improved reading speed for readers with and without dyslexia [11], and chromatic schemes improved aesthetic ratings without reducing retention [5], but no reviewed study directly compared a moderate-contrast warm scheme with black-on-white over a sustained reading session while measuring speed, comprehension, and fatigue together. The question therefore remains open, and the experiment proposed above is designed to close it.

In general, the evidence does not support the view that maximum contrast alone always produces the optimal reading experience. Contrast is better understood as a necessary baseline, after which designers should weigh polarity, colour selection, environmental lighting, user characteristics, and the intended reading task. Allowing users to customize these settings remains one of the most practical means of accommodating the wide variation observed between individuals and usage contexts.

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